

The Batman Function: A Generalized Framework for Graph Construction

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1 *Where it all started*

In 2011, a mathematical graph of the Batman logo went viral online, showing how equations could be combined to recreate something as visually iconic as the Batman symbol. What fascinated me most was that it wasn't created using a single elegant equation, but rather through a careful combination of mathematical ideas, symmetry, and graphing techniques to reverse-engineer an image. It became one of the earliest examples of mathematics being used not just for solving problems, but for creating visual art. Seeing how math could be transformed into something so creative inspired me to explore the idea further, and so I decided to experiment and develop my own method to draw the Batman logo through mathematics

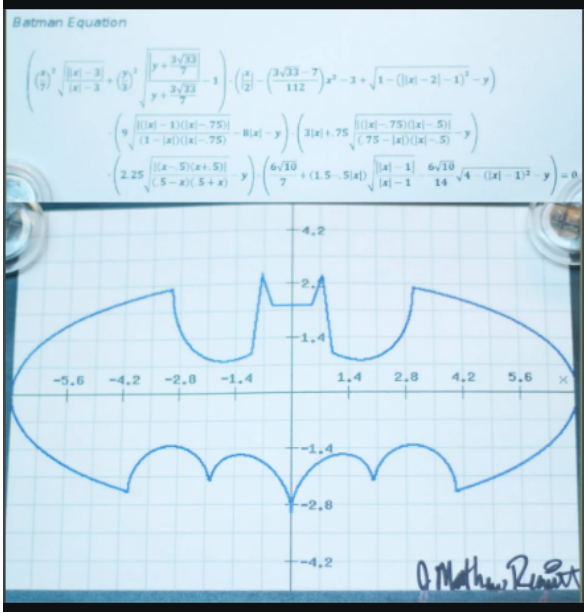


Figure 1: The original Batman graph from over 15 Years ago!

2 Intentions

The objective of this work is to construct a generalized framework that allows multiple functions to be combined into a single mathematical expression. In doing so, it becomes possible to recreate graphical structures such as the Batman logo on the Cartesian plane while respecting the fundamental constraints of a mathematical function. The best way to understand the construction is to follow the development step by step and build intuition for how the mechanism behaves.

3 Introduction(1)

This is a function intended to basically start from 0 and reach 1, and remain 1, for the parameter still increasing.

If any part of the explanation feels unclear, the accompanying Desmos visualization may help build intuition for how the function behaves.

4 how the function is built

We define "L" and it is intended to go from 0 to 1 and then terminate (return 0) once L exceeds 1, this can be easily defined:

$$f(L) = L - L \cdot \text{floor}(L)$$

the term " $\text{floor}(L)$ " will become "1" and terminate the function as soon as L reaches 1 although we don't want the value of that term to exceed 1 and hence we assign signum function to it and we now have:

$$f(L) = L - L \cdot \text{sgn}(\text{floor}(L))$$

So, once L exceeds 1, it will automatically terminate at 0.

Now suppose that we want the function to activate at 1 and terminate at 2. A simple modification gives:

$$f(L) = (L - 1) - (L - 1) \cdot \text{sgn}(\text{floor}(\frac{L}{2}))$$

so now say $L = 1.19$, the function will give 0.19 and eventually when reach 2, the floor function will become 1 and then terminate the whole function to zero, once " L " exceeds 2. from the above two cases we can come to the conclusion that, if we want the function to activate at " n " and hence terminate at " $n + 1$ " we can get a generalized version given below:

$$f(L) = (L - (n - 1)) - (L - (n - 1)) \cdot \text{sgn}(\text{floor}(\frac{L}{n}))$$

but we also have to realize that we do not want the function to fully terminate, we want it to be "1", so we add +1 and set it to be zero until the function reaches " $n + 1$ ":

$$f(L) = (L - (n - 1)) - (L - (n - 1)) \cdot \text{sgn}(\text{floor}(\frac{L}{n})) + (1) \cdot (\text{sgn}(\text{floor}(\frac{L}{n})))$$

on simplification we get:

$$f(L) = (L - (n - 1)) - (L - (n - 1) - 1) \cdot \text{sgn}(\text{floor}(\frac{L}{n}))$$

so now when you run the equation, you will see that when the value of L reaches the assigned value " n ", the function will start generating the values from 0 to 1 as the values of L go from $n - 1$ to n

An important observation is that before L reaches $n - 1$, the function produces unintended values. Ideally, we want the output to remain 0 in this interval, so we need a new multiplied parameter that stays zero until L reaches " $n - 1$ " and that function can be:

$$\text{sgn}(\text{floor}(\frac{L + 1}{n}))$$

now we simply multiply this term by the expression we found above

$$f(L) = ((L - (n - 1)) - (L - (n - 1) - 1) \cdot \text{sgn}(\text{floor}(\frac{L}{n}))) \cdot \text{sgn}(\text{floor}(\frac{L + 1}{n}))$$

Although the resulting expression appears complicated, its behavior is fairly intuitive when visualized. To better understand its action, we now examine its graph. In the interactive experience below:

NOTE

-all the blue texts are clickable links from here on.

[A.Desmos.Presentation](#)

5 Introduction(2)

So in the pages, I had explained about the function, which was given by:

$$f(L) = ((L - (n - 1)) - (L - (n - 1) - 1) \cdot \text{sgn}(\text{floor}(\frac{L}{n}))) \cdot \text{sgn}(\text{floor}(\frac{L+1}{n}))$$

Plotting the function over values of L gives the following behavior

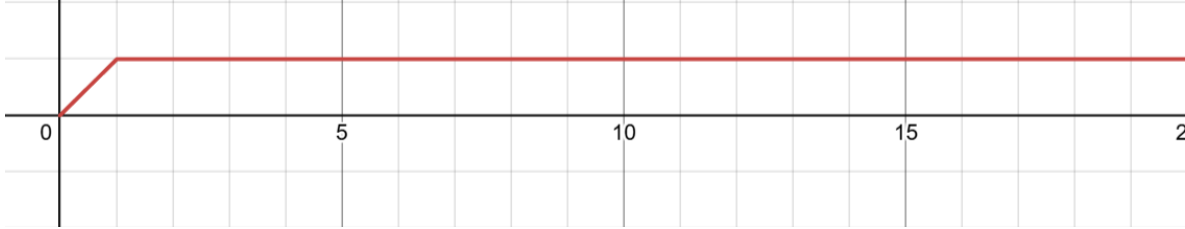


Figure 2: here $n = 1$

As shown in Figure 2, the function increases over the interval $L = 0$ to $L = 1$, as expected. To further illustrate the behavior, consider the same construction $n = 5$:



Figure 3: here $n = 5$

This demonstrates that for any value of n , the function activates at $n - 1$, reaches its maximum at n , and then stabilizes.

However, our goal is not a gradual increase to 1, but rather a discrete transition to 1. To convert this gradual increase into a sharp transition, we apply the ceiling function, basically which rounds off any real value to the closest integer higher than it

Our function now becomes:

$$f(L) = \text{ceil}(((L - (n - 1)) - (L - (n - 1) - 1) \cdot \text{sgn}(\text{floor}(\frac{L}{n}))) \cdot \text{sgn}(\text{floor}(\frac{L+1}{n})))$$

and now we get the graph:

well, now there is one last thing left..an end point,we do want the function to become zero again.

so we now introduce a new parameter " n_2 " and just for convenience sake, we will now call " n " as " n_1 "

so basically we want the function to return again 0,for any/all values exceeding " n_2 "i feel like by now the intuition is probably their...we can multiply the following term with our function:

$$\text{sgn}(\text{floor}(\frac{n_2}{L}))$$

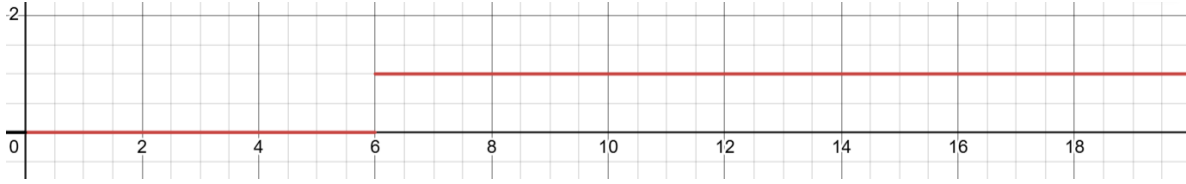


Figure 4: here $n = 5$

well, here, this function stays 1, until L exceeds the value of n_2 so after multiplying this term by our function, we get:

$$\text{ceil}(((L - (n_1 - 1)) - (L - (n_1 - 1) - 1) \cdot \text{sgn}(\text{floor}(\frac{L}{n_1}))) \cdot \text{sgn}(\text{floor}(\frac{L+1}{n_1})))(\text{sgn}(\text{floor}(\frac{n_2}{L})))$$

well, you realize until now, for any value of n_1 we take, (n_1 being the starting point) But clearly you see that the function starts (becomes 1) at $n_1 + 1$. so to simply fix this, we have replaced the $n_1 - 1$ term(s) with n_1

and we thus get:

$$\text{ceil} \left(L - (n_1) - (L - (n_1) - l) \left(\text{sgn} \left(\text{floor} \left(\frac{L}{n_1} \right) \right) \right) \right) \left(\text{sgn} \left(\text{floor} \left(\frac{(L+1)}{n_1} \right) \right) \right) \left(\text{sgn} \left(\text{floor} \left(\frac{n_2}{L} \right) \right) \right)$$

note: the following function is indeed a little more well shown, for visual clarity

At first glance the expression may appear cumbersome; however, its structure becomes clearer through experimentation, well with me showing you another screenshot of the function, An interactive Desmos implementation is provided to allow experimentation with the parameters

A brief note on the parameter l : its role is comparatively simple and is best understood through experimentation in the Desmos implementation.

[*Another.Desmos.Presentation*](#)

6 Input Variables

well, so now that you have checked out the function for yourself, Having established the construction, we can now examine how the function behaves under different parameter choices.

from now on, we can call our function as: $A_1(L, n_1, n_2, l)$ which, is of course given by:

$$A_1(L, n_1, n_2, l) = \text{ceil} \left(L - (n_1) - (L - (n_1) - l) \left(\text{sgn} \left(\text{floor} \left(\frac{L}{n_1} \right) \right) \right) \right) \left(\text{sgn} \left(\text{floor} \left(\frac{(L+1)}{n_1} \right) \right) \right) \left(\text{sgn} \left(\text{floor} \left(\frac{n_2}{L} \right) \right) \right)$$

The function is now fully determined by four variables: L, n_1, n_2 and l . For graphical purposes, we take $L = x$, since the function is plotted over X -axis. In most cases, $l = 1$, unless a vertical shift is desired

Readers are encouraged to experiment with different parameter values to observe how the function changes

the remaining 2 variables " n_1, n_2 " they are basically the starting and ending points...

A concrete example makes the mechanism significantly easier to understand.

7 Example

Consider the following example: suppose we want a function that behaves like $\sin(x)$ over the interval $x \in [1, 4]$, and like $\log(x)$ over $x \in [4, 7]$.

well so let's look at our Desmos interface, which is where we are going to perform this task!..

Figure 5 shows the equation in Desmos

$$A_1(L, n_1, n_2, l) = \text{ceil}\left(L - (n_1) - (L - (n_1) - l) \left(\text{sgn}\left(\text{floor}\left(\frac{L}{n_1}\right)\right) \right) \left(\text{sgn}\left(\text{floor}\left(\frac{(L+1)}{n_1}\right)\right) \right) \left(\text{sgn}\left(\text{floor}\left(\frac{n_2}{L}\right)\right) \right) \{L > 0\}\right)$$

Figure 5: Our function in Desmos

7.1 STEP-1

simply write the following on Desmos:

$$f(x) = \text{Sin}(x)$$

7.2 STEP-2

since we want the *Sine* function only from " $x = 1$ " to " $x = 4$ " we input the following in our function A_1

$$L = x$$

$$n_1 = 1$$

$$n_2 = 4$$

The parameter l may be adjusted depending on the desired vertical displacement
similarly do the same for $\log(x)$ term and add it to your $f(x)$

Our function now:

$$f(x) = A_1(x, 1, 4, 1) \cdot \text{Sin}(x) + A_1(x, 4, 7, 1) \cdot \log(x)$$

and The resulting graph produced in Desmos is shown below::

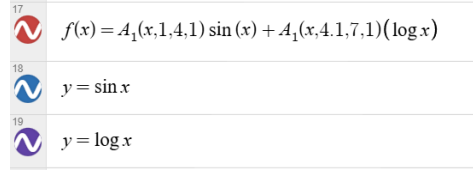


Figure 6: Equation shown on the desmos interface.

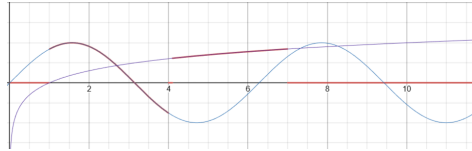


Figure 7: The graph desmos shows us

While the previous example is intentionally simple, a more complex application is the reconstruction of the Batman logo (figure 7)

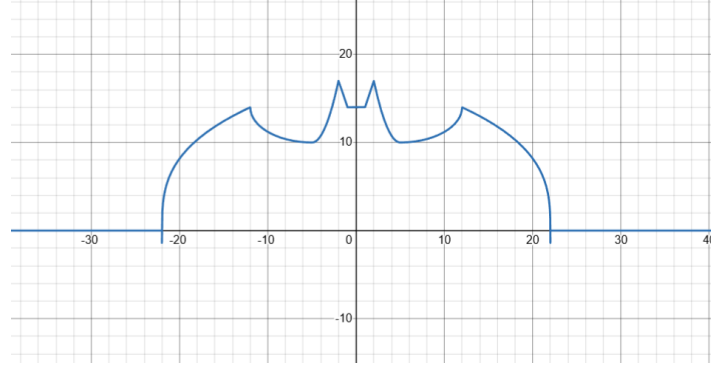


Figure 8: BATMAN

Equation:

$$\begin{aligned}
y = & \\
& 14 \left(A_1(|x|, 0, 1, 1) \right) \\
& + \\
& (3(|x| - 1) + 14) A_1(|x|, 1, 2, 1) \\
& + \\
& \left(\frac{7}{3^2} (-|x| + 5)^2 + 10 \right) A_1(|x|, 2, 5, 1) \\
& + \\
& \left(- \left| \left(\left(1 - \frac{(|x| - 5)^2}{7^2} \right) 4^2 \right) \right|^{\frac{1}{2}} + 14 \right) A_1(|x|, 5, 12, 1) \\
& + \\
& \left(- \left(\frac{(|x| - 22) 14^3}{10} \right) \right)^{\frac{1}{3}} A_1(|x|, 12, 22.01, 1)
\end{aligned}$$

For readability, the expression has been split across multiple lines.,
and for reference where these equations came from, The underlying component equations are shown below:

Note: these equations [The *Raw.Batman.Equations*] can be found using brute force, by thinking of appropriate methods..

Raw.Batman.Equations

For clarity and proper attribution: the raw Batman equations used here were sourced externally (can be sourced easily from the internet) and are not my original work. My contribution lies in the construction of the generalized function framework used to combine them into a unified

8 Limitations

8.1 How a function is defined, in reflection to what is simply not possible.

Since we are not just defining a relation, rather a function, there are limitations.

if you notice in both examples above for every " x " there is only one " y " value defined...a basic principle on how functions work,

in the same way we cannot complete the full batman logo as, we would then end up having 2 values of " y " for one " x "

a good way to think about it is, that our default function is a string, but has infinite length between any 2 points taken at a time, at the same time it can only be bent in a single way at a point, in fact the string can even be cut, but once again cannot exist at 2 places at once.